

Universidad Politécnica de Madrid - Escuela Técnica Superior de Ingenieros de Telecomunicación (ETSIT)



Design, Development and Evaluation of an Explainable Artificial Intelligence Pipeline for Predicting Cybersickness Escalation Using Native Virtual Reality

Early prediction of cybersickness escalation using native VR eye- and head-tracking signals



Master in Science in Neurotechnology

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Why is cybersickness still a major challenge?

VIRTUAL REALITY IS TRANSFORMING MANY FIELDS



Healthcare



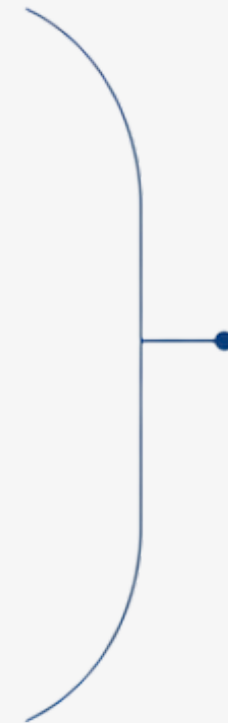
Education



Industry



Entertainment



CYBERSICKNESS LIMITS THE VR EXPERIENCE



Nausea



Dizziness



Disorientation



Visual fatigue



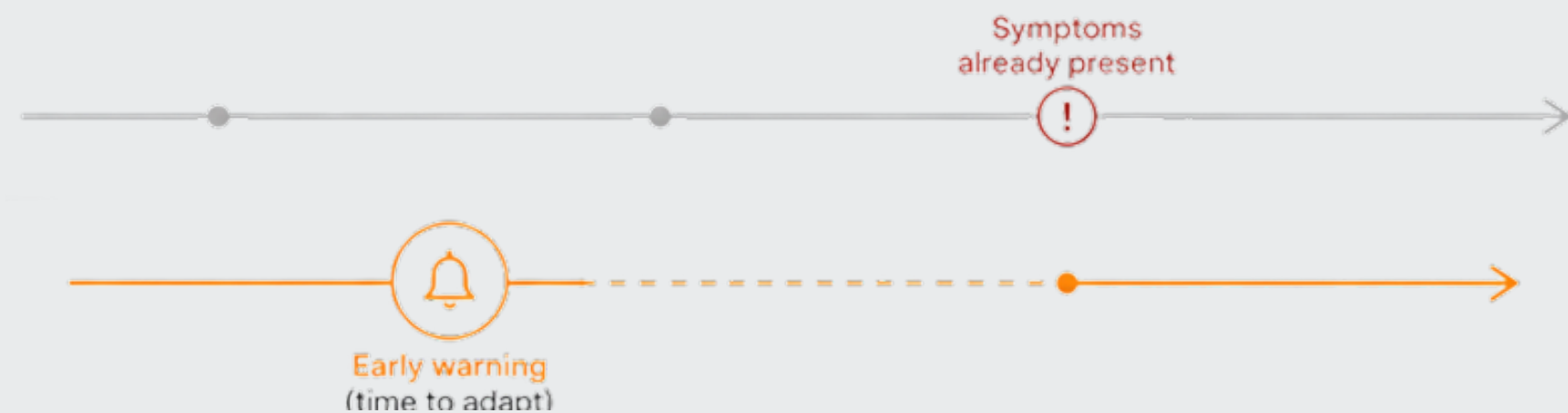
CURRENT SYSTEMS

React only after symptoms appear



THIS THESIS

Predict before symptom escalation



Research goals

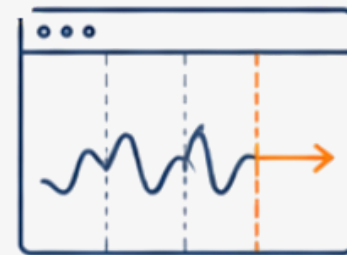
This thesis pursues **four main objectives**:



1

Understand the data

Explore native VR eye- and head-tracking signals and cybersickness annotations.



2

Build an early-warning pipeline

Design a pipeline that anticipates cybersickness escalation before symptoms appear.



3

Compare ML and DL models

Evaluate a wide range of machine learning and deep learning approaches.



4

Explain forecasting

Apply Explainable AI techniques to identify the most relevant signals and provide insights.

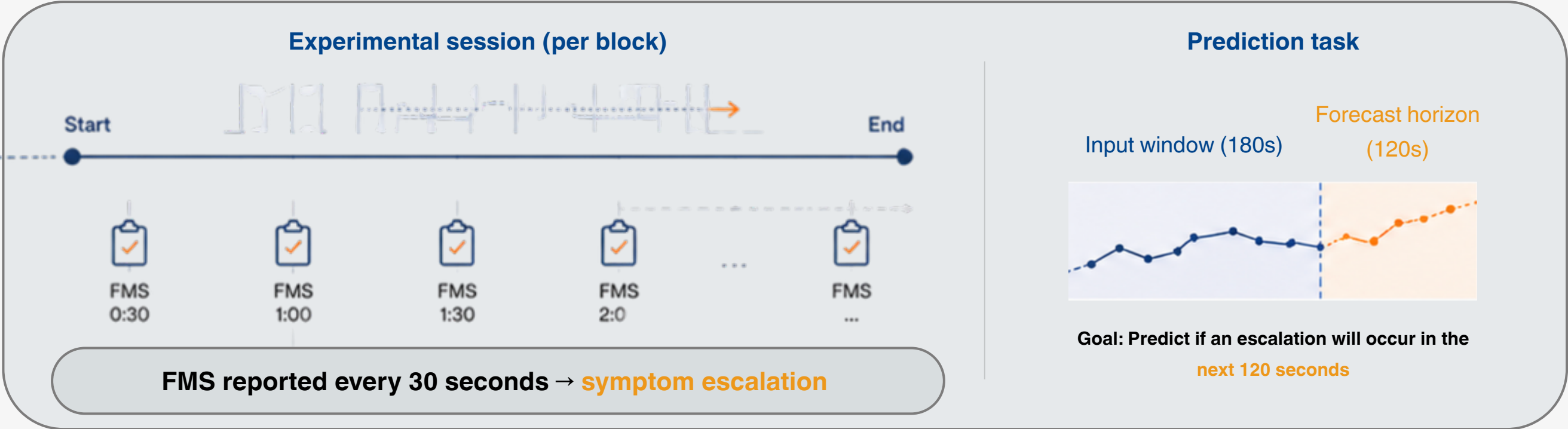


Ultimately, the goal is to develop an explainable AI system capable of **predicting cybersickness escalation before it affects the user experience.**

Dataset & Experimental Protocol

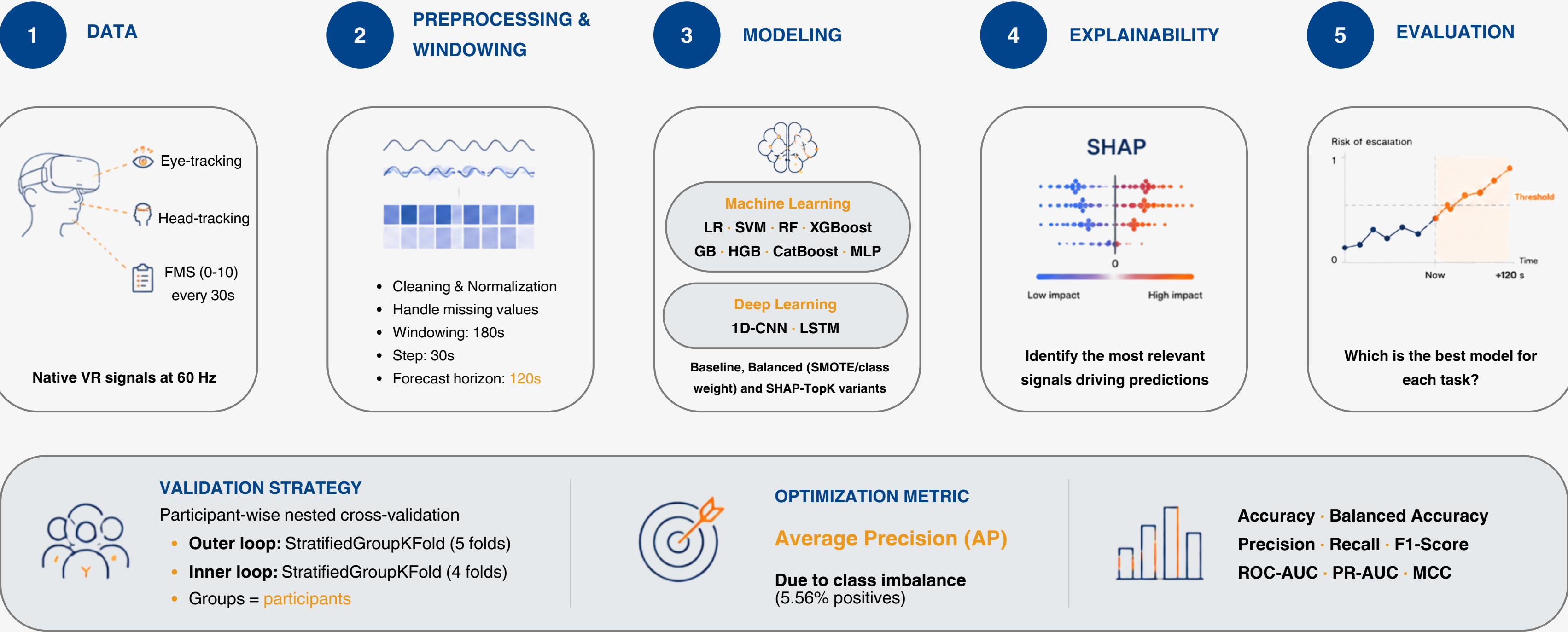
Data collected in the MazeStick / RelaxVR study at University of Missouri

-  **Participants**
60 participants
-  **Environment**
Immersive VR navigation (MazeStick / RelaxVR)
-  **Signals**
Native VR eye-tracking and head-tracking at 60 Hz
-  **Cybersickness measure**
FMS questionnaire every 30 seconds (0-10)



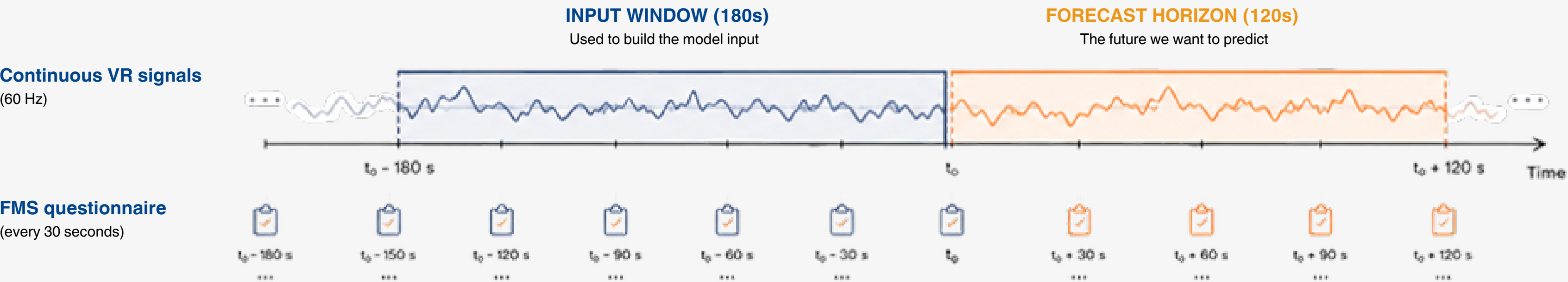
Methodology at a glance

From raw VR signals to early prediction of cybersickness escalation



Forecasting, not just detection

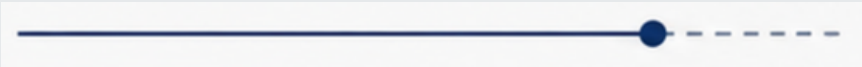
Goal: predict if an escalation will occur in the next 120 seconds.



At each time t_0 , we use the past 180 s of signals (input window) to predict whether an escalation will occur in the next 120 s (forecast horizon). → Output : 1 (escalation) or 0 (no escalation)
Target label = Escalation within next 120 s



DETECTION (what most systems do)
Identify that cybersickness is present after symptoms have already appeared.



Too late for intervention

VS.



FORECASTING (our approach)
Anticipate an escalation before symptoms increase.



Time to adapt and intervene



This ability to anticipate cybersickness escalation **is the key contribution of this thesis.**
It enables early warnings and proactive interventions.

Design, Development and Evaluation of an Explainable AI Pipeline for Predicting Cybersickness Escalation using Native VR

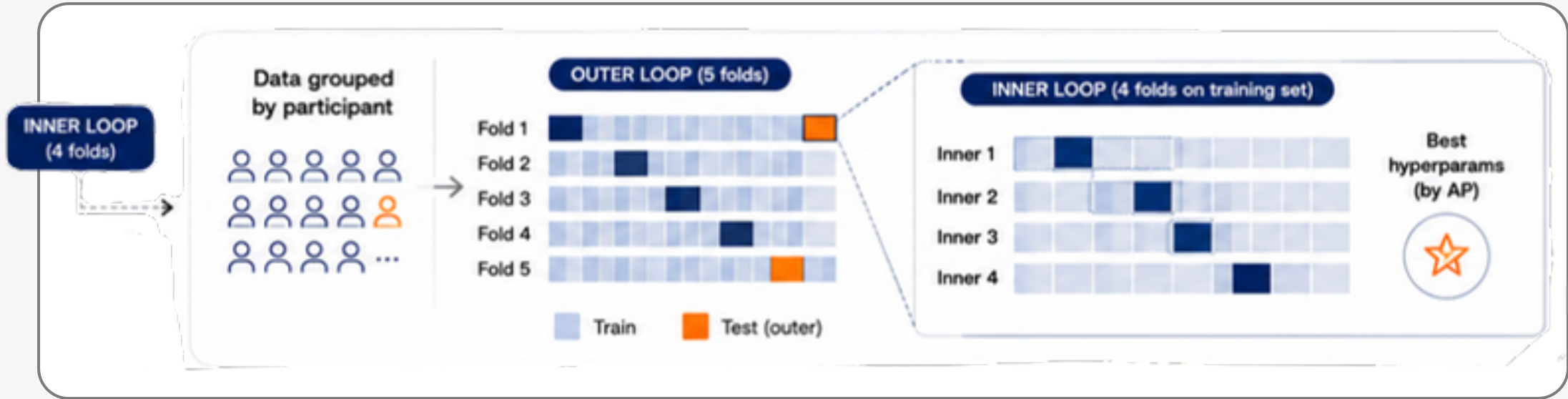
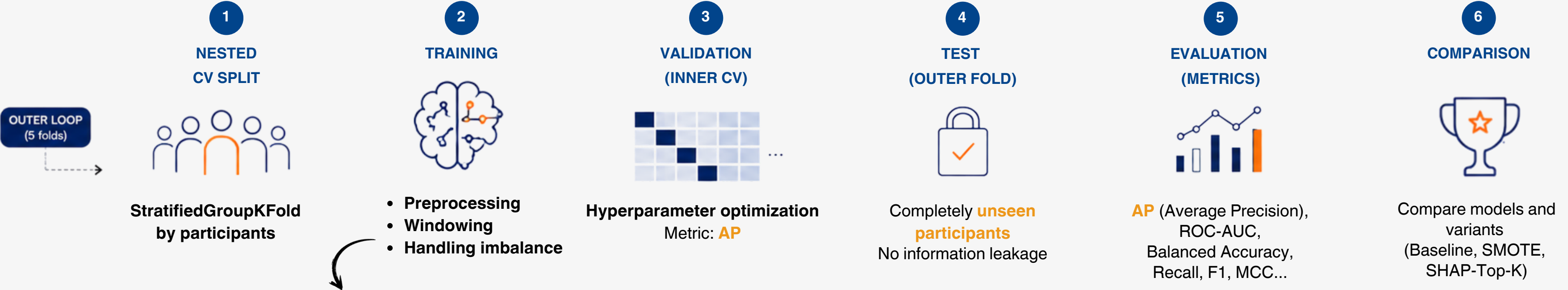
06

Machine learning pipeline

Participant-aware nested cross-validation and evaluation

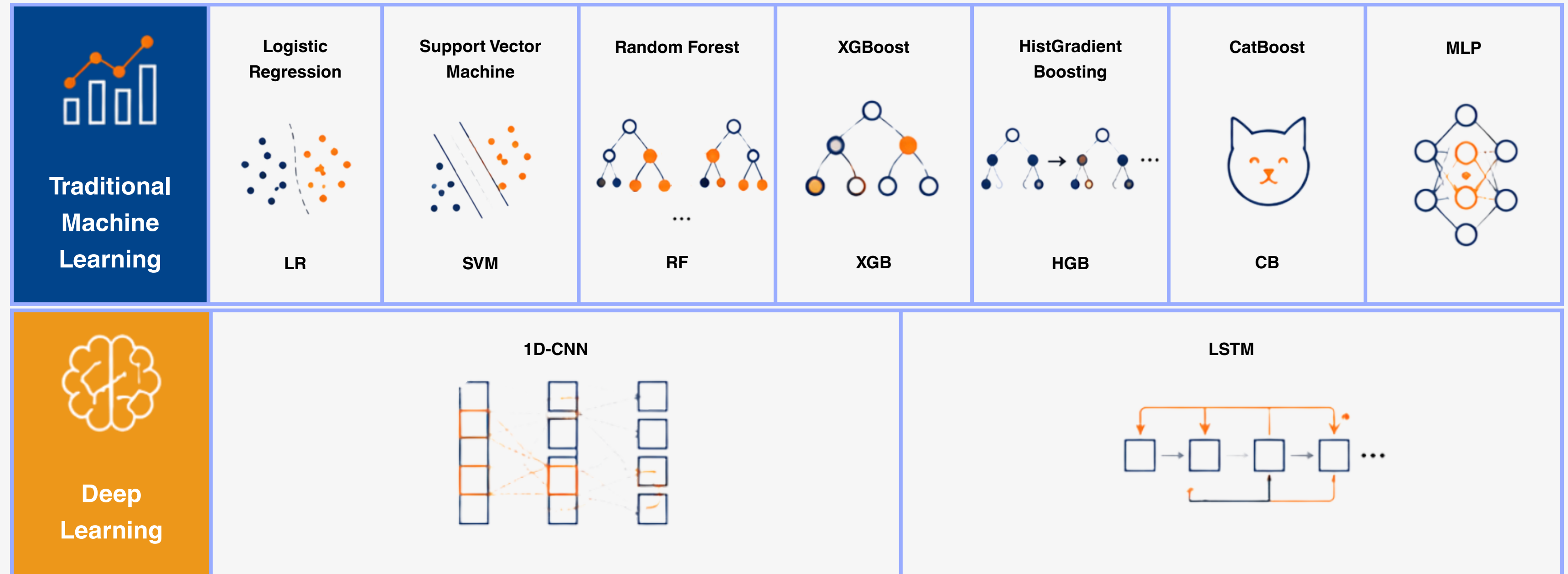
 Train

 Test (held-out)



Models evaluated

From interpretable linear models to deep temporal architectures.



All 22 models evaluated under the same rigorous **nested cross-validation** framework with **participant-aware** splits.

Understanding what drives predictions

Using SHAP values to explain model decisions and identify the most relevant signals

SHAP (SHapley Additive exPlanations):
Python library for Explainable AI



SHAP

Quantifies the contribution of each feature to individual predictions.



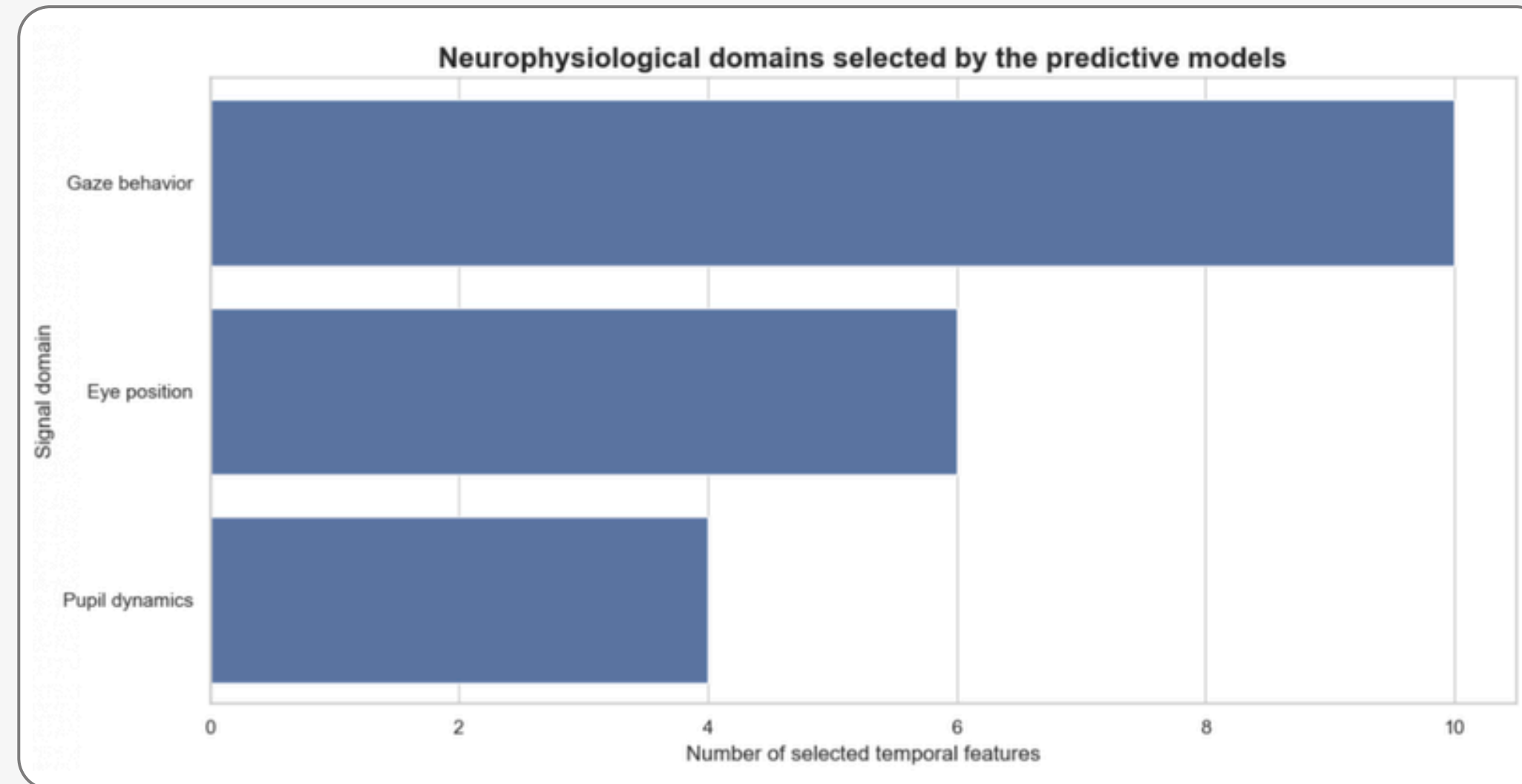
Feature selection

We select the top 20 features according to their mean |SHAP| value.



Interpretability

Reveals which signals increase or decrease the risk of escalation.



SHAP help us **trust** the model, identify **informative signals**, and provide **neurophysiological insights**.

Best performing models

Evaluation on the independent test set (participant-aware split)

Model (Variant)	bal_acc	F1	pr_auc	roc_auc	mcc	rank_score	recall (1)
1D-CNN	0.69	0.20	0.15	0.78	0.19	0.32	0.65
LSTM	0.66	0.23	0.17	0.76	0.20	0.32	0.45
MLP - SMOTE + FS	0.64	0.20	0.12	0.80	0.17	0.29	0.45
XGB - Baseline	0.62	0.18	0.14	0.69	0.13	0.28	0.40
LR - SMOTE + FS	0.57	0.12	0.08	0.69	0.07	0.22	0.50

$rank_score = 0.4 \cdot pr_auc + 0.3 \cdot bal_acc + 0.3 \cdot mcc$



1D-CNN

Best model for early detection of escalation events (highest recall & bal_acc).



LSTM

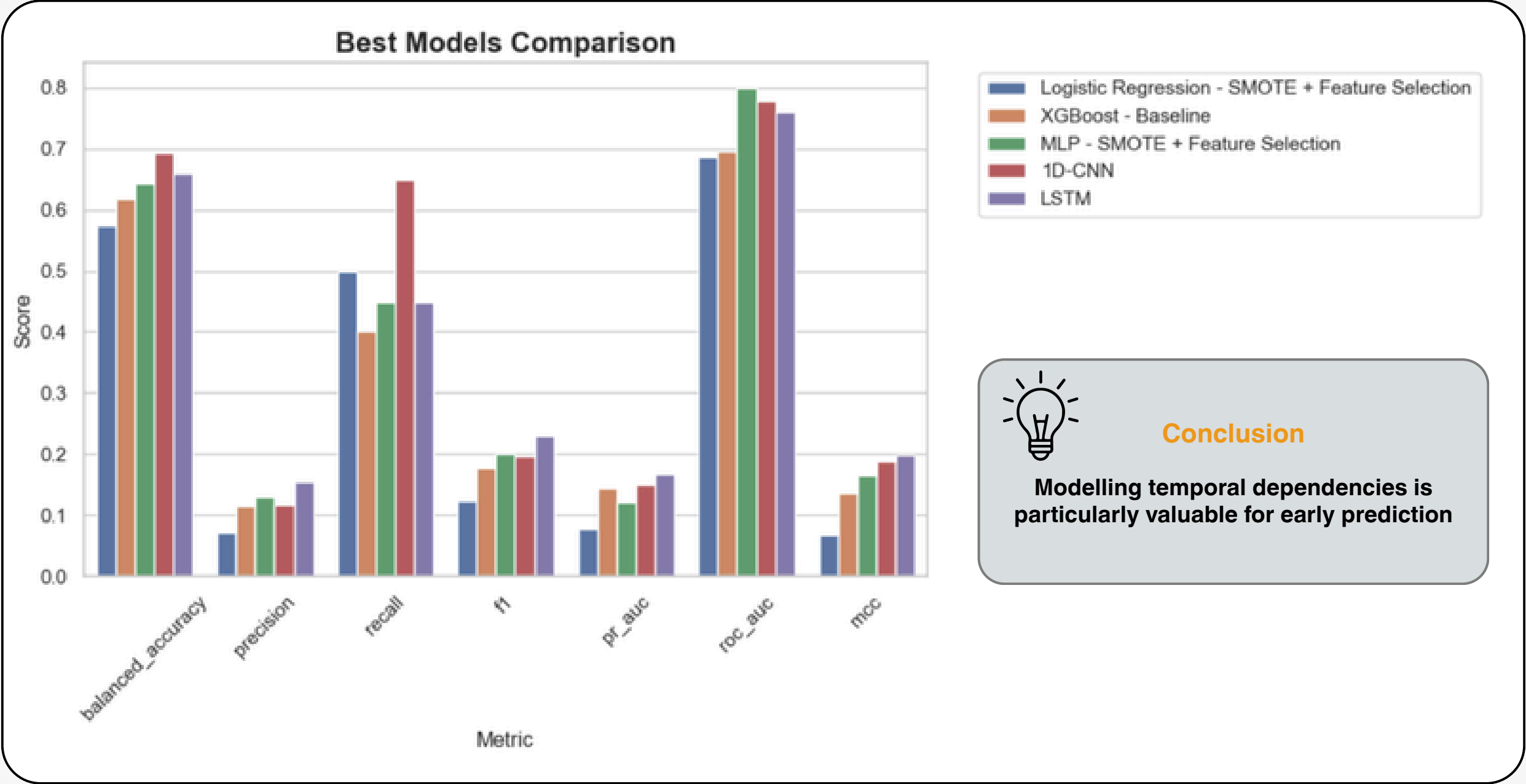
Best overall trade-off between precision and recall (highest pr_auc, f1 and mcc).

- Deep Learning outperformed classical ML.
- Temporal architectures better captured pre-event dynamics.
- Multiple models achieved competitive performance.



Best performing models

Evaluation on the independent test set (participant-aware split)



Neurophysiological signature before escalation

Temporal evolution of the most informative VR signals during the last 3 seconds before symptom escalation.



✓ **Eye-tracking signals change before escalation.**

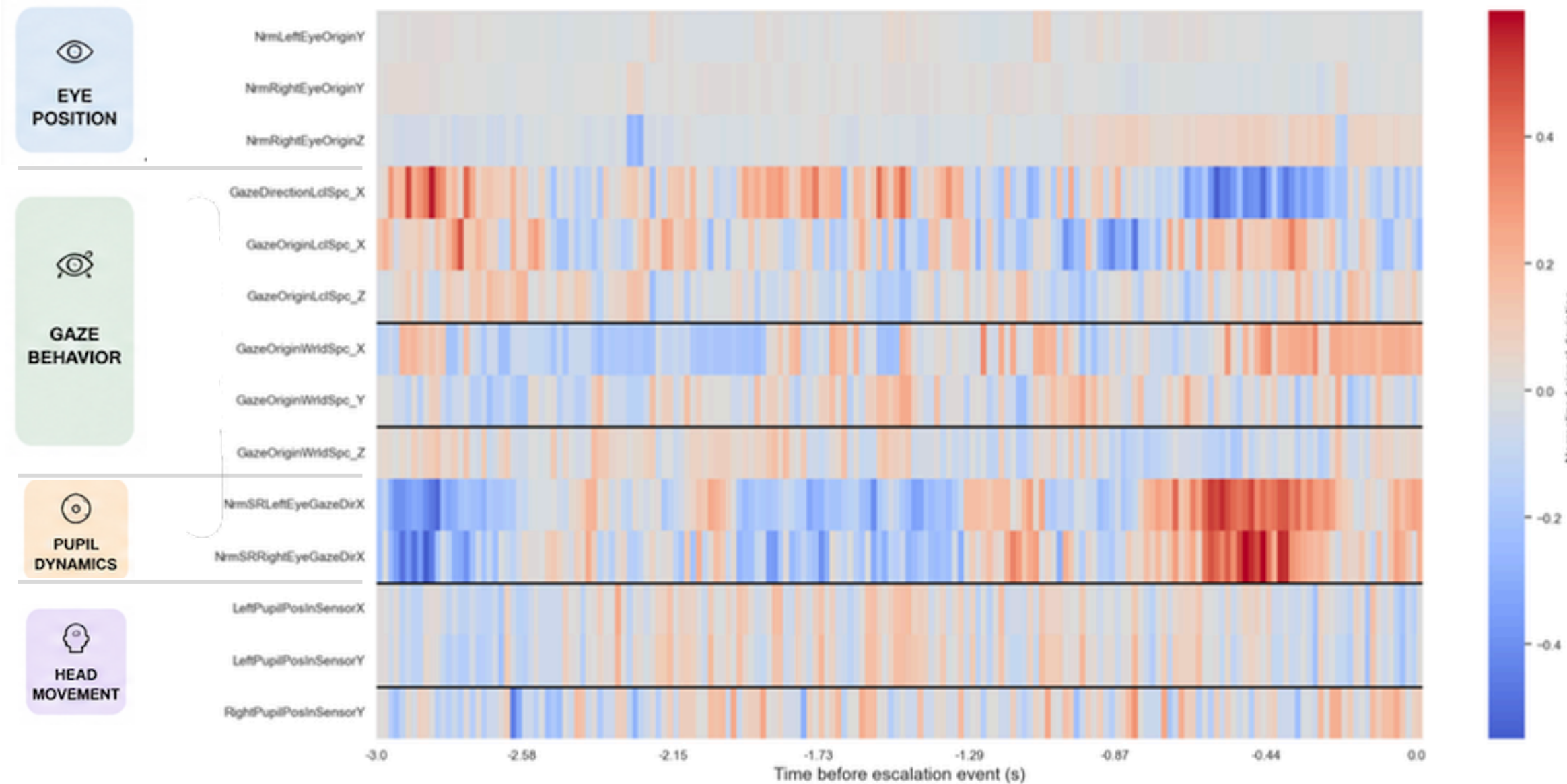
Systematic deviations are visible in multiple gaze-related metrics.

✓ **Gaze direction exhibits the strongest temporal variations.**

Both eyes show marked shifts in the last second before the event.

✓ **Multiple signals evolve together before the event.**

Coordinated physiological changes reflect increasing visual instability.



Consistent pre-event physiological changes support the feasibility of early-warning prediction.

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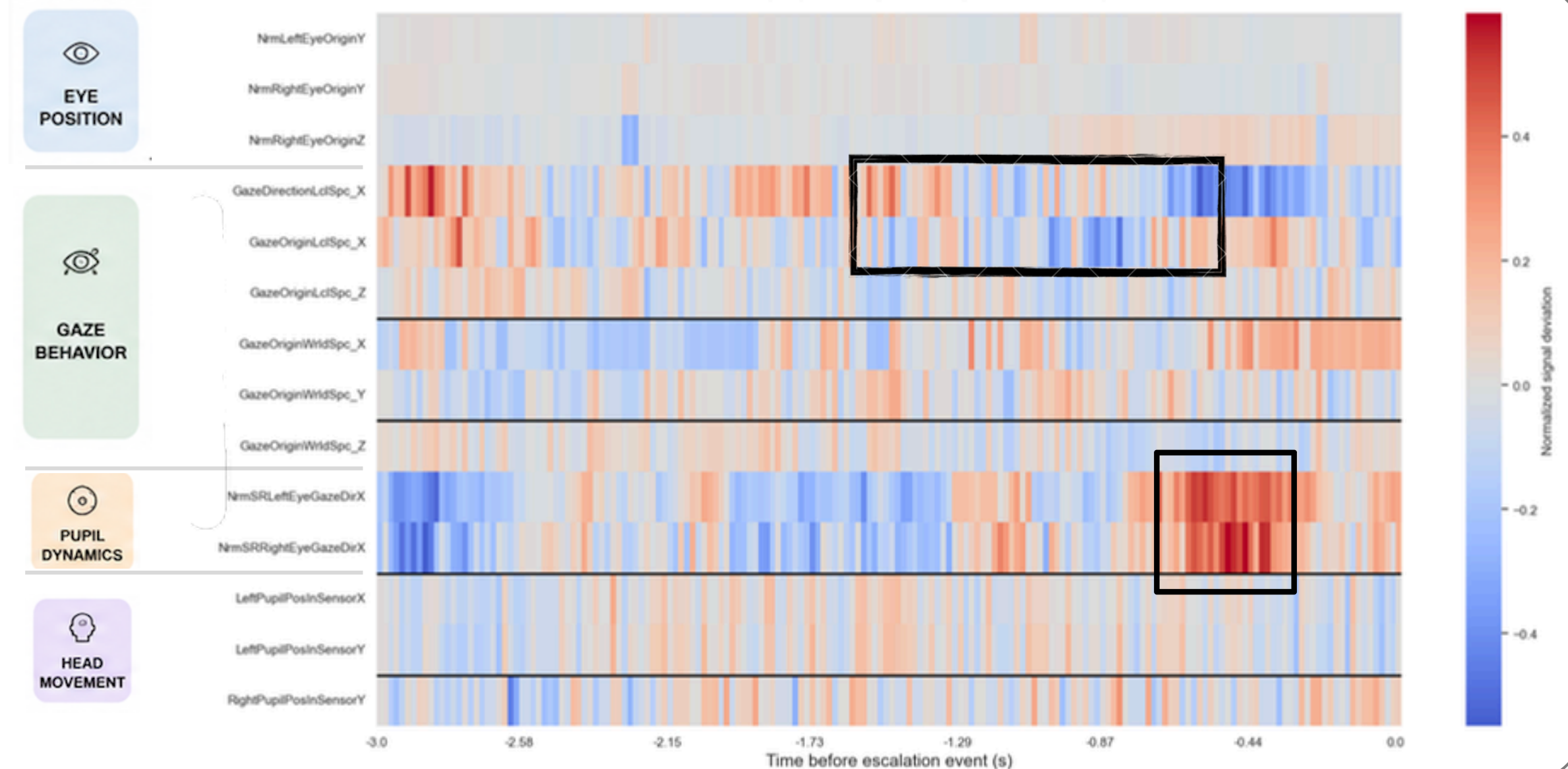
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Individual early-warning case study

Example: participant 4

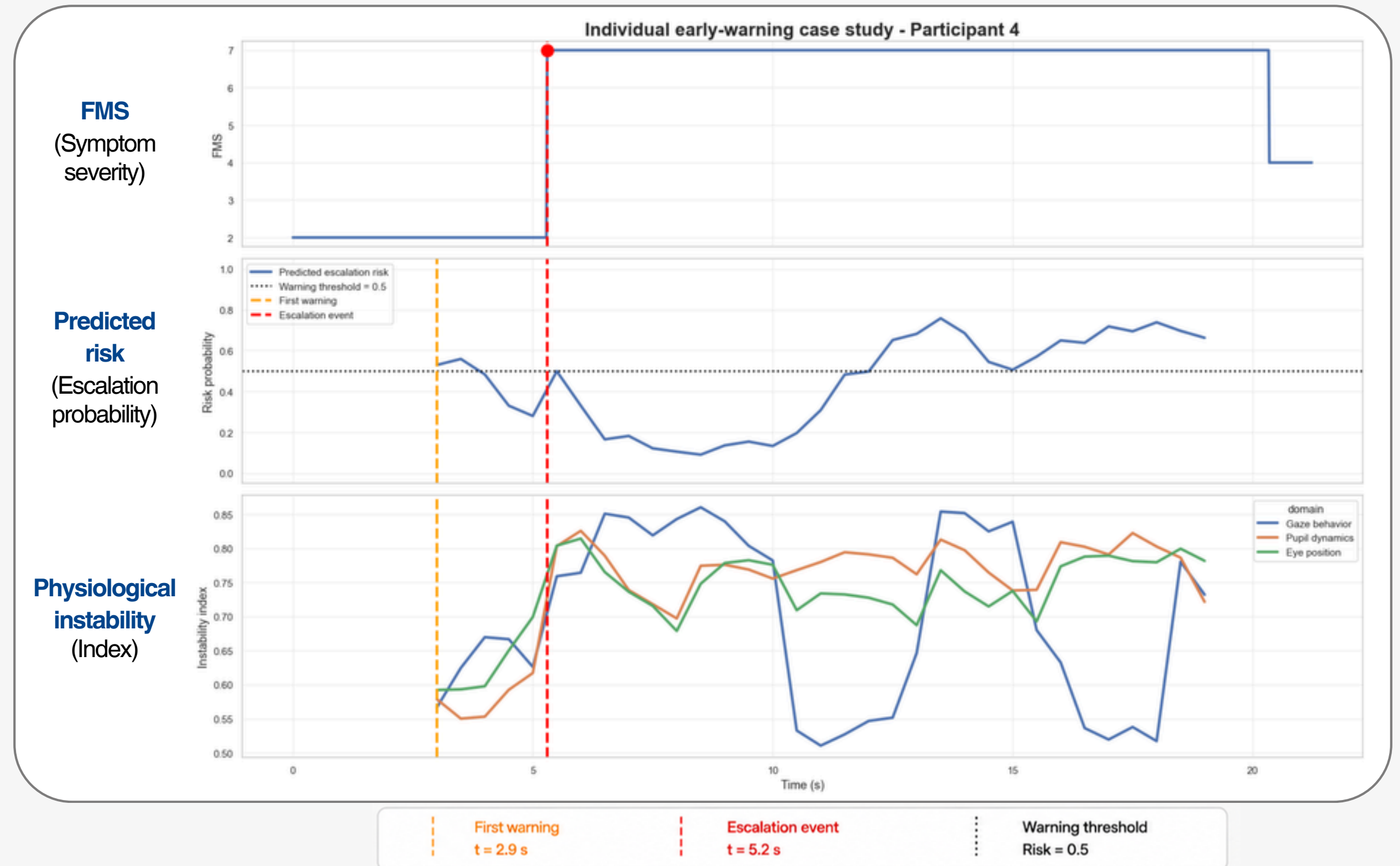
 **The model issues an early warning before symptom escalation occurs**

 **Early-warning**
The risk exceeds the threshold ~2.3 seconds before escalation

 **Escalation event**
The FMS score increases from 2 to 7 at time 5.2 s.

 **Physiological changes**
Gaze behavior, pupil dynamics and eye position show increased instability before the event.

 **The system anticipates escalation, enabling adaptive interventions before discomfort increases.**



Main contributions

Key advances of this work



1

Native VR signals

Demonstration that native VR eye-tracking and head-tracking signals contain sufficient information to anticipate cybersickness escalation.



2

Explainable AI pipeline

Design and implementation of an end-to-end pipeline including preprocessing, feature engineering, modeling, evaluation and interpretability.



3

Early-warning forecasting framework

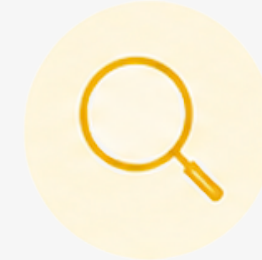
Formulation of the problem as a forecasting task to predict cybersickness escalation before it occurs, enabling proactive interventions.



4

State-of-the-art performance

Deep learning models (1D-CNN and LSTM) outperformed traditional ML approaches, achieving strong results in early detection of escalation events.



5

Neurophysiological insights

Identification of robust pre-escalation patterns in gaze behavior, eye position and pupil dynamics that characterize increasing cybersickness risk.



6

Practical relevance and impacts

Demonstration of the feasibility of anticipating discomfort in real time, opening the door to adaptive VR systems that improve user comfort and safety.



This work **demonstrates that native VR signals contain sufficient information to anticipate cybersickness escalation**, laying the foundation for **adaptive** and **personalized** VR experiences

Conclusions & Future Work

From predictive models to real impact in immersive experiences

What are the limitations?

-  **Dataset size.**
Only 60 participants were available, so larger cohorts are needed to further validate the models.
-  **Single VR scenario.**
The models were trained on MazeStick/RelaxVR. Generalization to other VR applications should be evaluated.
-  **Limited physiological modalities.**
Only native VR eye- and head-tracking signals were used.
-  **Offline evaluation.**
The pipeline was evaluated offline and has not yet been deployed in real-time VR systems.
-  **Model architectures.**
Evaluate transformer-based and hybrid temporal architectures for improved long-term temporal modeling.

Where to go next

- 1 Real-time deployment**
Optimize the pipeline for low-latency inference and online operation.
- 2 Personalization**
Adapt models to individual susceptibility and behavior over time.
- 3 Multi-modal integration**
Incorporate additional physiological signals (HRV, GSR, EEG) to improve robustness.
- 4 Broader validation**
Test the system in diverse VR scenarios, populations and longer sessions.
- 5 Adaptive interventions**
Design and evaluate mitigation strategies triggered by predicted risk.



Publication

This work will be submitted for publication in a peer-reviewed journal in the field of virtual reality, human-computer interaction and neurotechnology.



Impact

These results contribute to the development of proactive, adaptive VR systems that enhance user comfort, reduce discomfort and expand the possibilities of immersive technologies.

Thank you!

This work bring us closer to **safer** and more **immersive** virtual experiences through **early detection**.



Questions?

